

PROJECT PLAN DOCUMENT

Project number	28
Project Title	Fleet Management
Document	ProjectPlan
Creation date	January21,2025
Created By	Akshitha
Client	AkhilReddyAnthamgari

Problem Statement

The topic for this project is the challenge of optimizing employee transportation by designing efficient routes in situations with many pickup and drop-offs. The problem is minimizing travel time and traveling cost by adhering to other constraints like vehicle capacities and schedules of drivers and passengers. Again, there are two of the most common cases, for instance, sending occupants from various points destined-off and a single destination point (Many-to-One), or managing lengthy commutes with multiple start and end points (Many-to-Many).

Team Members

1. **Akshitha** : Geocode employee addresses and Route Visualisation
2. **Aryanil** : Optimization Algorithm
3. **Aviral** : Integration of the algorithms and Communication
4. **Satvik** : Preprocessing
5. **Sherley** : Data Acquisition and Import

Team Communication

- Meet every Saturday at 2PM in Workspace.

Development Environment:

IDE:

Visual Studio Code

Programming Languages:

Frontend: TypeScript with React for the user interface

Backend: Next.js

Google APIs for Route Optimization:

- **Google Maps Platform:**
 - Google Maps Directions API: For calculating optimal routes between locations
 - Google Maps Distance Matrix API: For calculating travel time and distance between multiple origins and destinations
 - Google Maps Geocoding API: For converting addresses into geographic coordinates

Visualization Libraries:

- Leaflet: For displaying interactive maps

Collaboration Tools:

- Communication: Gmail, Whatsapp
- Documentation: Microsoft Word, Ms Excel, GitHub

Optimization Techniques and Algorithms:

- Traveling Salesperson Problem (TSP) Algorithms: For single-route optimizations
- Vehicle Routing Problem (VRP) Algorithms: For multi-vehicle optimizations
- Heuristic Algorithms: Genetic Algorithms or Simulated Annealing for complex scenarios

Additional Considerations:

- Google Maps Traffic Data: For real-time traffic information and dynamic routing

Milestone Schedule

Milestone	Due Date	Release	Deliverable?
Create draft requirements	23/01/25	R1	No
Finalize requirements	26/01/25	R1	Yes
Project setup and planning	02/02/25	R1	No
Initial documentation	31/01/25	R1	Yes
Google Maps API integration	10/03/25	R1	Yes
Google Geocoding API implementation for employee locations	15/03/25	R1	No
Basic routing algorithm using Google Directions API	13/03/25	R1	Yes
Many-to-One routing optimization	17/03/25	R1	Yes
Simple user interface for data input	1/03/25	R2	Yes
Many-to-Many routing capability	6/04/25	R2	Yes
Final testing and bug fixes	10/04/25	R2	No
Documentation completion	11/04/25	R2	Yes
Final Delivery	15/04/25	R2	Yes